

Advanced Machine Learning

DATA 642

Lecture 1 — Motivation and Introduction to Vector Calculus

*Notes and figures are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth

Motivation

Many algorithms in machine learning optimize an objective function with respect to a set of desired model parameters that control how well a model explains the data: Finding good parameters can be phrased as an optimization problem. Examples include:

- Linear regression, where we look at curve-fitting problems and optimize linear weight parameters to maximize the likelihood.
- Neural-network auto-encoders for dimensionality reduction and data compression, where the parameters are the weights and biases of each layer, and where we minimize a reconstruction error by repeated application of the chain rule.
- Gaussian mixture models for modeling data distributions, where we optimize the location and shape parameters of each mixture component to maximize the likelihood of the model. (used in clustering, anomaly detection)

Central to those examples is the notion of the objective function which needs to be optimized in order to estimate the best set of parameters and achieve a desirable performance associated to the machine learning task. We denote a function by $J(\boldsymbol{\theta}) : \mathbb{R}^l \rightarrow \mathbb{R}$.

Example: Linear Regression

The goal in regression is to estimate a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{l+1}$ given a training set $(y_n, \mathbf{x}_n)$ for $n = 1, \dots, N$. The prediction model is given by

$$\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x_1 + \hat{\theta}_2 x_2 + \dots + \hat{\theta}_l x_l = \hat{\boldsymbol{\theta}}^\top \mathbf{x}_n \quad (1)$$

and we estimate the parameter vector by minimizing the following cost function

$$J(\boldsymbol{\theta}) = \sum_{n=1}^N (y_n - \hat{\boldsymbol{\theta}}^\top \mathbf{x}_n)^2. \quad (2)$$

To minimize $J(\boldsymbol{\theta})$ we need to solve $\frac{\partial J}{\partial \boldsymbol{\theta}} = \mathbf{0}(*).$ Using matrix notation, equation $(*)$ becomes

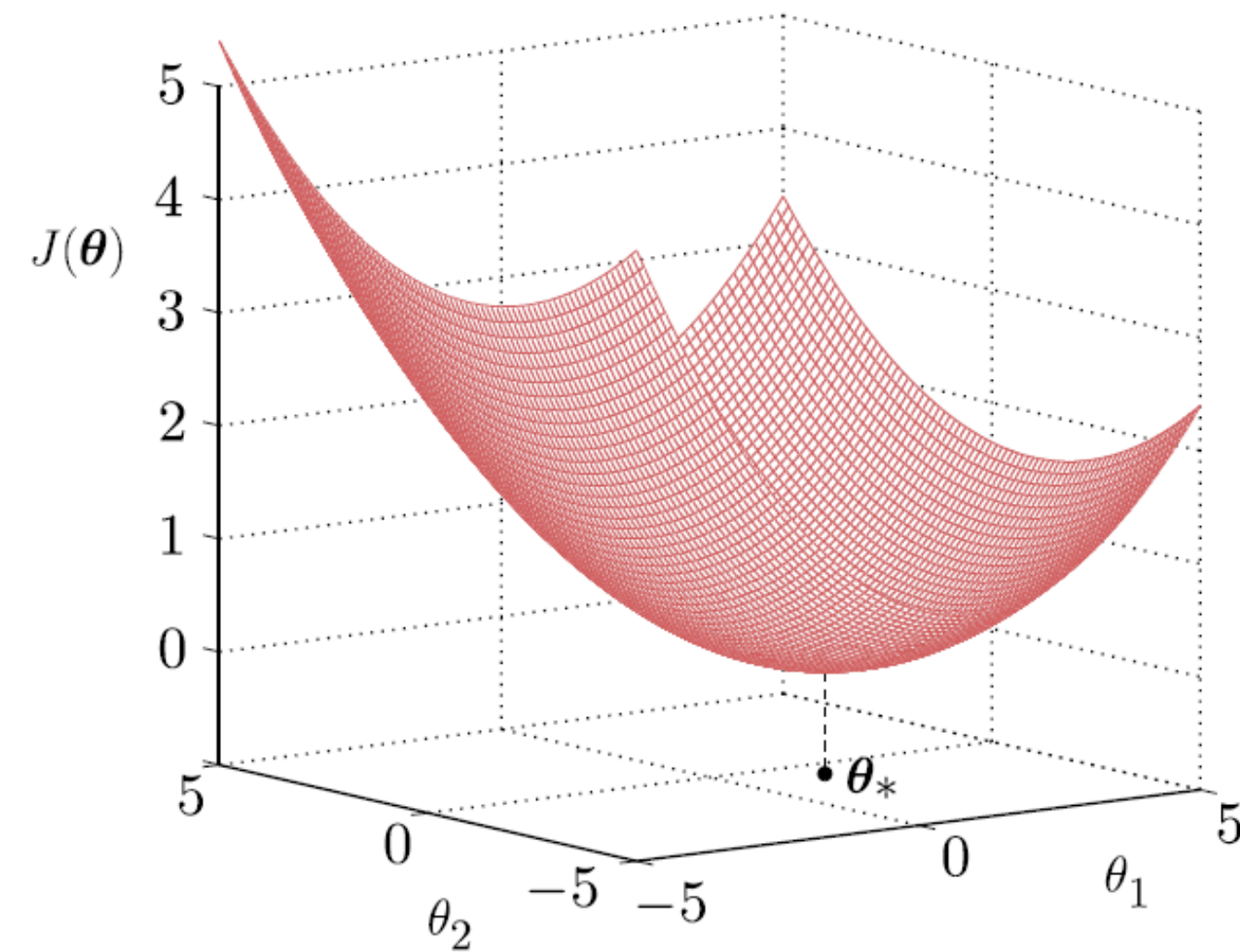
$$(\mathbf{X}^\top \mathbf{X}) \hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y}, \quad (3)$$

where $\mathbf{y} = [y_1, y_2, \dots, y_N]^\top$. Therefore, the least squares estimate is given by

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (4)$$

assuming that $(\mathbf{X}^\top \mathbf{X})^{-1}$ exists.

The next figure illustrates the squared error loss function for $l = 2$. This objective function has a unique minimum. As it can be seen, this function is convex and thus, a minimum is a global minimum.



Concepts Map

The following chart includes the main mathematical concepts that are mainly used in machine learning.

